Program: **Diploma in Food Processing Technology**	
Course Code: **3422**	Course Title: **Elements of Food Engineering**
Semester: **3**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4 (L:3 T:1 P: 0)**	Periods per semester: **60**

Course Objectives:

1. To impart knowledge of various physical, aero and hydrodynamic properties of food materials
2. To learn about the frictional, thermal, electrical and electromagnetic properties of food materials
3. To impart knowledge on rheological properties its classification etc.

Course Prerequisites:

Topic	Program/Course Name
Basic Knowledge of Applied Physics	Applied physics I&II

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain physical, aero and hydrodynamic properties of food materials.	17	Understanding
CO2	Discuss theoretical knowledge of frictional properties of food materials	15	Understanding
CO3	Distinguish thermal, electrical and electromagnetic properties of the food	15	Understanding
CO4	Explain rheological properties related to qualitative and quantitative attributes of food.	13	Understanding

CO–POMapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Stronglymapped, 2-Moderatelymapped, 1-Weaklymapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Explain physical, aero and hydrodynamic properties of food materials.**		
M1.01	Explain the concept of various physical properties such as volume, porosity.	6	Understanding
M1.02	Describe the various hydrodynamic properties such as, density, viscosity etc.	5	Understanding
M1.03	State the various aerodynamic properties such as drag coefficient and terminal velocity	6	Understanding
Contents: Definition and concepts of volume, density (apparent, material, particle, bulk, true) porosity, viscosity, drag coefficient, terminal velocity.			
CO2	**Discuss theoretical knowledge of frictional properties of food materials**		
M2.01	Describe the laws of friction	5	Understanding
M2.02	Review kinetic friction, static friction, coefficient of friction, rolling resistance	5	Understanding
M2.03	Define angle of repose, angle of internal friction	4	Understanding
	Series Test –I	1	
Contents: Laws of friction, Definition, concept of – kinetic friction, static friction, coefficient of friction, rolling resistance, Definition and concept of angle of repose, angle of internal friction.			

CO3	**Distinguish thermal, electrical and electromagnetic properties of the food**		
M3.01	Describe thermal conductivity	5	Understanding
M3.02	Define specific heat, measurement of specific heat	5	Understanding
M3.03	State thermal diffusivity, boiling point, freezing point	5	Understanding
Contents: Thermal conductivity-Definition, concept, Specific heat- Definition, concept, measurement, Concepts and definition of thermal diffusivity, boiling point, freezing point.			
CO4	**Explain rheological properties related to qualitative and quantitative attributes of food.**		
M4.01	Describe the concept and definitions of rheology	4	Understanding
M4.02	Classify the rheology of food materials	4	Understanding
M4.03	Illustrate the physical states of matter, classic ideal materials	4	Understanding
	Series Test –I	1	
Contents: Rheology- Definition, concept and classification, Physical states of matter, classic ideal materials.			

TEXT/REFERENCE

T/R	Book Title/Author
R1	Mohensenin N N, Physical properties of plants animal materials, Gordon and Breach publishers, New York, 1980
R2	Rao M A , Rizvi S S H,Azim K Datta and Jasim Ahmed, Engineering properties of foods 4th Ed., CRC Press, 2014
R3	K M Sahay and K K Singh, Unit operations of Agricultural Processing, Second edition, Vikas Publishing India Pvt Ltd, 2004
R4	Ignacio Arana, Physical Properties of Foods Novel Measurement Techniques and Applications, 1st edition, CRC Press, 2016

R5	Singhal,O.P. and Samuel,D.V.K, “Engineering Properties of Biological Materials”. Saroj Prakasan, Allahabad 2003
R6	Peleg, M.and BagelayE.B., “Physical properties of foods”. AVI publishing Co. USA 1983.

Online Resources

Sl.No	WebsiteLink
1	https://www.khanacademy.org/science/physics/thermodynamics/specific-heat-and-heat-transfer/a/what-is-thermal-conductivity
2	http://ecoursesonline.iasri.res.in/mod/page/view.php?id=1013
3	https://www.nature.com/articles/eye2017267
4	https://acta.fih.upt.ro/pdf/2022-3/ACTA-2022-3-15.pdf